Rotation-Aware Adaptor via Equivariance-Constrained Consistency Regularization for Remote Sensing Detection

Chuling Wen, Weijie Liang, Jian Lu, Chen Xu, and Yuru Zou

***Abstract*—Remote sensing has become indispensable in applications such as urban planning, maritime surveillance, and disaster management, yet accurate object detection remains challenging due to arbitrary orientations, dense layouts, and large intraclass variations. Existing oriented detectors have made progress through rotation-sensitive heads and feature-alignment mechanisms, but they still handle rotation primarily in an implicit manner, lacking explicit geometric priors and often exhibiting inconsistency between feature representations and final predictions. To address these limitations, we design a *parameter-efficient modular adaptor* readily applicable to diverse detectors with only a small learnable-parameter overhead. Specifically, we introduce a rotation-aware multi-mother convolution (RAMMC), which explicitly embeds rotation as a structural inductive bias: a multibranch adaptor enhances rotation-aware feature construction, while two complementary regularization terms enforce feature-level equivariance and prediction-level consistency, ensuring coherent and geometry-aligned detection behavior. The design is inspired by rotation equivariance and invariance theory. We provide a conditional theoretical analysis showing that the group-pooling branch of RAMMC realizes a Reynolds averaging operator over the discrete rotation group, and we derive an implementation-aware equivariance-defect bound that accounts for finite kernel rotation, spatial gating, and orientation-attention residuals. Extensive experiments on diverse detectors, such as Faster R-CNN, RoI Transformer, and Oriented R-CNN implemented in the MMRotate toolbox, as well as YOLO models implemented under the Ultralytics framework, confirm that our adaptor consistently improves performance. Notably, it achieves 84.32 mean Average Precision (mAP) on DOTA-v1.0, 47.31 mAP on FAIR1M-v1, and 52.28 mAP on FAIR1M-v2.0, surpassing strong baselines. These results demonstrate that explicitly modeling rotation consistency through a parameter-efficient adaptor with mathematically grounded rotation priors and consistency constraints provides an effective and general solution for remote sensing object detection.**

***Index Terms*—Consistency regularization, oriented detection, parameter-efficient adaptor, remote sensing, rotation equivariance.**

Received 28 December 2025; revised 29 April 2026 and 23 June 2026; accepted 11 August 2026. Date of publication 14 August 2026; date of current version 25 August 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62372302, Grant U21A20455, Grant 12571569, and Grant 12326619; in part by the Natural Science Foundation of Guangdong Province of China under Grant 2023A1515011691 and Grant 2024A1515011913; and in part by Shenzhen Basis Research Project of China under Grant JCYJ20210324094006017. *(Corresponding author: Jian Lu.)*

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Digital Object Identifier 10.1109/TGRS.2026.3723812

I. Introduction

OBJECT detection in remote sensing imagery has long been a central yet highly challenging problem, driven by critical applications such as maritime surveillance, aircraft monitoring, urban planning, and disaster response [1], [2], [3], [4]. Unlike natural images, aerial and satellite imagery are typically acquired under far more complex conditions, often featuring ultra-high-resolution scenes where many targets occupy only a few pixels and appear with arbitrary orientations amid cluttered backgrounds. In addition, sensor-dependent degradations such as SAR speckle, haze, and illumination imbalance further increase intraclass variation and geometric uncertainty, posing fundamental challenges to rotation-robust detection [5], [6], [7].

A critical bottleneck lies in the use of horizontal bounding boxes (HBBs) for object localization, as this approach fails to accurately enclose densely distributed and arbitrarily rotated targets. To address this, the community began to adopt oriented bounding boxes (OBBs), which introduce an additional rotational degree of freedom for tighter and more accurate localization. This shift was catalyzed by the release of DOTA [8], the first large-scale remote sensing dataset with OBB annotations, enabling the development of tailored methods for rotation-aware detection.

Building on this representation, Ding et al. [9] proposed RoI Transformer, which introduced a learnable module to align features with rotated RoIs, significantly improving detection in dense scenes. Xu et al. [10] developed gliding Vvrtex, which sidesteps unstable angle regression by refining HBBs through vertex sliding. Yang et al. [11] presented R3Det, a refined single-stage detector that iteratively aligns features and predictions. Xie et al. [12] proposed KFIoU loss, which reformulates intersection over union (IoU) as a Gaussian distribution matching problem, thereby alleviating regression instability and enhancing robustness in complex remote sensing scenarios. In addition, recent works on feature interaction and cross-branch cooperation, such as mutual-assistance learning [13], further strengthen representation learning by enabling collaborative feature enhancement across different detection

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pathways. Moreover, many other excellent methods have also been developed to further advance rotation-aware detection in remote sensing imagery [14], [15], [16], [17], [18], [19].

However, despite substantial gains driven by architectural innovations and parameter scaling, modern detectors remain fundamentally limited in their ability to treat geometric transformations, particularly rotation, as *first-class priors*. In most existing approaches, rotation sensitivity is still modeled in an implicit manner—typically through data augmentation, heuristic feature alignment, or auxiliary angle regression, rather than being embedded as a structural modeling principle within the network and its optimization objectives. This may result in inconsistent behaviors between intermediate feature representations and final predictions, and reduced robustness under extreme rotations and dense layouts [8], [11], [12], [15], [20], [21]. This suggests that the core challenge is not merely how to design rotation-aware operators, but how to explicitly model rotation consistency throughout feature extraction, prediction, and training.

To bridge this gap, we propose a unified rotation-aware design that treats rotation consistency as an explicit principle throughout the detection process. Rather than relying on an isolated operator modification, our method combines a parameter-efficient adaptor-based feature design with dual-level consistency regularization, so that geometric priors are enforced in both intermediate representations and final predictions. To maintain compatibility with existing detectors, this formulation is instantiated through a parameter-efficient modular adaptor that can be inserted into standard backbones without altering the overall detection pipeline.

A qualitative comparison between the proposed method and the baseline is shown in Fig. 1. Concretely, as illustrated in Fig. 2, we introduce a parameter-efficient module, termed rotation-aware multi-mother convolution (RAMMC), to enhance detection accuracy under arbitrary object orientations. RAMMC is embedded at the block level of the backbone in parallel with the original block, and their outputs are fused via an attention mechanism. This design preserves the original feature extraction capability while injecting rotation-aware representations. Specifically, RAMMC employs a set of mother kernels, each of which generates multiple child kernels through parameter sharing, thereby explicitly encoding rotational patterns without a proportional increase in parameter complexity.

In particular, RAMMC acts as a structural rotation-aware adaptor that injects explicit geometric priors into the feature representation, addressing the lack of explicit rotation modeling in existing detectors. Meanwhile, the feature-level equivariance regularization and prediction-level consistency regularization terms reduce representation and output inconsistency under rotated inputs, thereby mitigating the aforementioned challenges. These components jointly enforce geometrically coherent behavior across the detection pipeline, improving robustness under extreme rotations and dense layouts.

Notably, as shown in Fig. 1, our model predicts bounding boxes that are often more compact and geometry-aligned than the baseline predictions. This indicates that the network does not simply replicate potentially noisy or biased labels, but instead learns rotation-aware representations that capture intrinsic object geometry, resulting in tighter boxes with less background redundancy. Such compactness has also been emphasized in related studies, e.g., Gaussian Wasserstein Distance loss for geometric alignment [22], distribution-based optimization of oriented boxes [23], as well as IoU-aware 3-D detection in autonomous driving [24], all highlighting that compact, geometry-aligned boxes improve robustness and reduce localization errors.

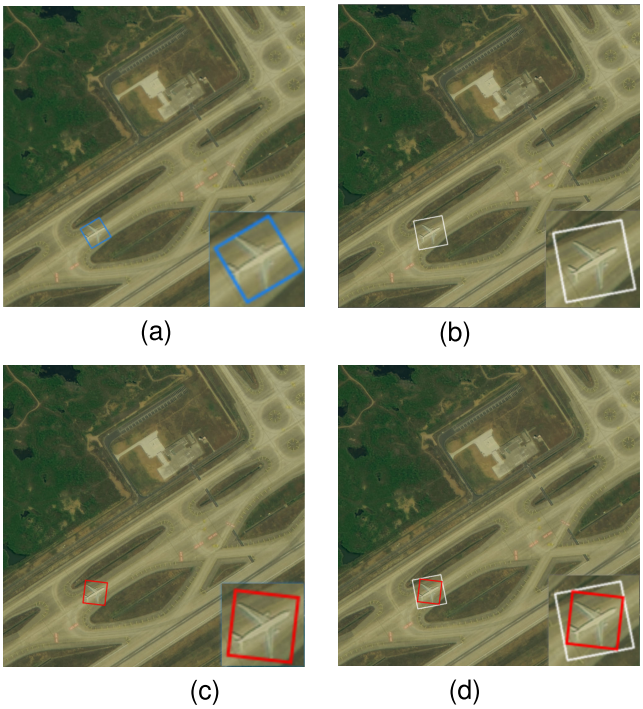


Fig. 1. Qualitative comparison of bounding box predictions on remote sensing images. (a) Ground-truth annotation. (b) Baseline prediction. (c) Our method. (d) Direct comparison between baseline and ours.

Extensive experiments on DOTA and FAIR1M benchmarks validate the effectiveness of our approach. In particular, our method demonstrates stronger robustness under extreme rotations, elongated aspect ratios, and densely packed targets scenarios where conventional detectors typically degrade.

Our Contributions

The main contributions of this article are as follows.

1) We propose a unified adaptor-based rotation-aware design for remote sensing object detection, in which a parameter-efficient modular adaptor (RAMMC) injects explicit geometric structure into existing detectors in a detector-compatible manner, without changing the overall detection pipeline.
2) We introduce a dual-level consistency regularization strategy that explicitly constrains both intermediate feature representations and final prediction outputs under geometric transformations, thereby enforcing rotation-consistent detection behavior.
3) We provide an implementation-aware theoretical analysis of the proposed design. In particular, we define the Reynolds operator induced by group pooling and derive a computable equivariance-defect bound, where

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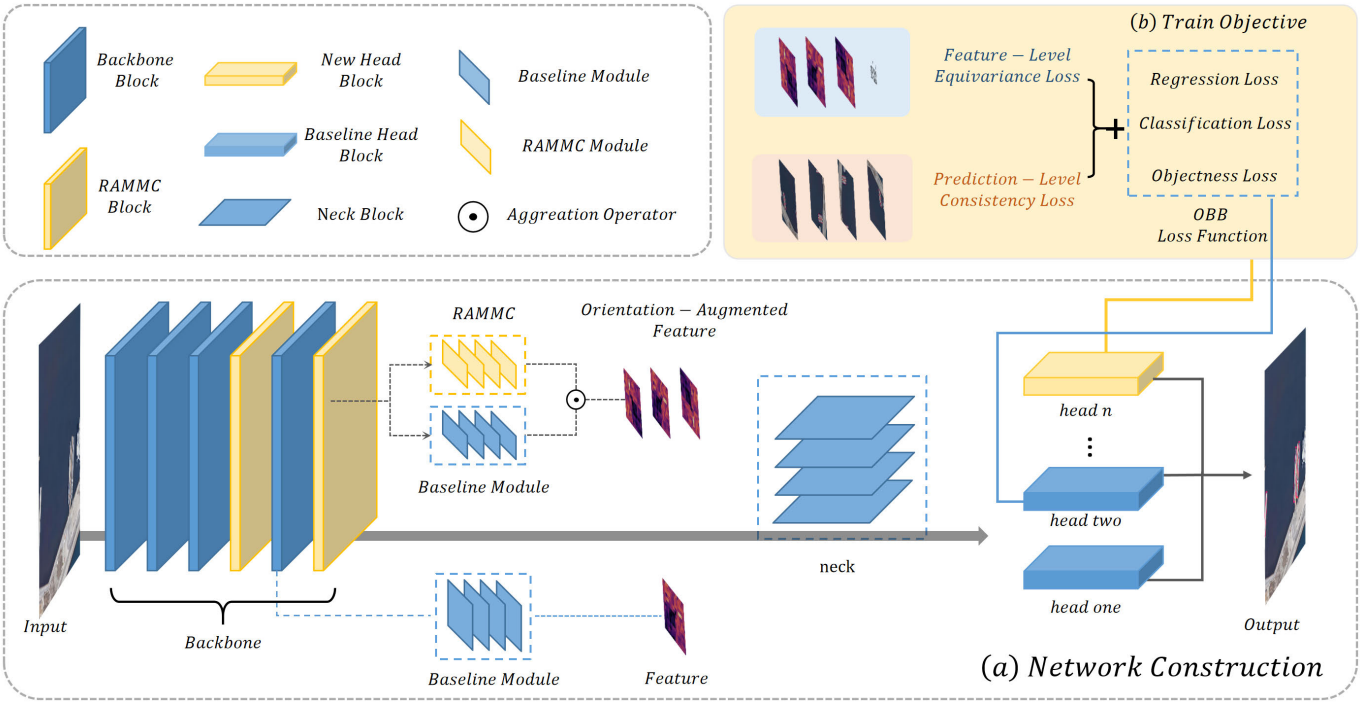


Fig. 2. Overall framework of the proposed rotation-aware detector. (a) Network construction. RAMMC is inserted into selected backbone stages as a parameter-efficient modular adaptor and aggregated with the corresponding baseline module to produce orientation-augmented features, which are then propagated to the neck and detection heads. (b) Training objective. On top of the original OBB loss, the proposed method further introduces a feature-level equivariance loss and a prediction-level consistency loss to enforce rotation-consistent learning.

the constants are related to the spectral norms of the implemented gating and attention modules.

4) We conduct extensive experiments on large-scale benchmarks. Our method achieves 84.32 mean average precision (mAP) on DOTA-v1.0, 47.31 mAP on FAIR1M-v1 [25], and 52.28 mAP on the extended FAIR1M-v2.0, consistently surpassing state-of-the-art baselines.

The remainder of this article is organized as follows. Section II reviews related work on remote sensing object detection and rotation-equivariant networks. Section III presents the proposed method, including adaptor design, dual regularization, and theoretical analysis. Section IV reports experimental results and ablation studies. Section V concludes this article and discusses potential future directions.

II. Related Work

A. Object Detection Paradigms in Remote Sensing

Early object detection methods in remote sensing relied on HBBs, which offer coarse localization but struggle with densely packed or arbitrarily oriented targets. To address this, the community shifted toward OBBs, which introduce rotational degrees of freedom for tighter fitting. This transition was catalyzed by the release of DOTA [8], which provided a large-scale benchmark for OBB-based detection.

Existing OBB-based detectors can be broadly categorized into three methodological streams.

1) Angle Regression and Iterative Refinement. Methods such as Gliding Vertex [10] and R3Det [11] improve angle prediction stability through cascaded refinements, while KFIoU loss [12] reformulates overlap estimation to stabilize optimization. These approaches largely focus on mitigating the nonlinearity of angle regression through loss function design or multistage pipelines.

2) Feature Alignment and Rotation-Aware Modules. RoI Transformer [9] aligns features with rotated proposals via learnable transformations, and subsequent works such as S2ANet [30] and ReDet [15] extend this by incorporating orientation-aware backbones and sampling strategies. These methods emphasize the importance of geometric alignment within the feature extraction process. Meanwhile, feature interaction and multibranch fusion mechanisms, such as mutual-assistance learning for object detection [13], also enhance representation learning through cross-branch cooperation. However, these methods are generally designed from an empirical architectural perspective, without explicitly modeling rotation as a structural geometric prior as in our RAMMC.

3) Modular Frameworks and Practical Deployment. Recent implementations such as MMRotate [21] provide modularized frameworks for building OBB detectors using standard backbones such as Faster R-CNN and YOLO, yet primarily rely on architectural heuristics or anchor-level modifications. While these systems improve usability and reproducibility, they often lack principled geometric modeling.

Beyond detection mechanisms, many efforts target task-specific challenges in remote sensing. These include compact and efficient designs for SAR [3], [4], contextual attention for ship detection [1], multimodal fusion [26], [27], and label-efficient learning strategies such as semi-supervised [29] and few-shot detection [28]. While effective in their respective

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domains, these methods generally operate orthogonally to the core problem of geometric representation.

Overall, existing detectors primarily encode rotation sensitivity either implicitly through data augmentation or indirectly via architectural heuristics. Very few works attempt to *explicitly embed geometric priors* or *enforce rotation-aware consistency* within the detection pipeline. This motivates our proposed method, which unifies feature-level inductive biases with mathematically grounded constraints to achieve robust rotation-aware detection.

B. Rotation Invariance and Equivariance

Rotation symmetry has been extensively studied to enhance robustness in recognition, detection, and restoration tasks, where objects and structures often appear under arbitrary orientations. Two complementary strategies have emerged: *equivariance*, which ensures predictable transformations under input rotations, and *invariance*, which suppresses orientation while retaining semantics.

Existing approaches can be broadly categorized into three methodological streams:

1) Group-Theoretic Formulations. Early attempts exploited weight sharing and rotation pooling [31], [32], later formalized as group-equivariant CNNs [32] and oriented response networks [33]. Steerable and harmonic filters [34], [35], [36] extended these ideas to continuous-angle steerability by tying kernels to irreducible group representations, while B-spline CNNs [37] generalized equivariance to Lie groups. These methods introduced mathematical rigor but were often constrained by discretization or computational overhead.

2) Spectral and Operator-Based Parameterizations. To improve expressiveness, spectral methods such as Fourier-series filter parametrization [38] alleviated aliasing while supporting strict equivariance, while PDO-based designs [39] provided discretization error bounds. More general frameworks extend equivariance to arbitrary Lie groups [40]. Building on this, rotation-equivariant proximal operators were embedded into deep unfolding to encode rotational priors with provable error bounds, further yielding rotation-equivariant encoders with implicit neural representations for arbitrary-scale super-resolution, demonstrating improved structural fidelity in low-level vision [41].

3) Relaxed Regularization and Applications. Strict equivariance, while theoretically appealing, is often broken in practice due to cropping, nonmaximum suppression, or oriented bounding box regression. To address this, invariant pooling [42], spatial transformer networks [43], and recent rotation-invariant graph models [44] aimed to suppress orientation sensitivity. In detection, equivariant backbones such as ReDet [15] and invariant heads such as RIDet [45] improved rotation robustness, while equivariant flows [46] and E(n)-equivariant diffusion [47] advanced generative modeling. Applications in medical imaging also highlight the benefits of joint scale- and rotation-equivariance [48]. A common trend is to relax hard constraints into differentiable regularizers or loss functions, balancing symmetry with flexibility.

The trajectory of rotation symmetry modeling has progressed from heuristic pooling and data augmentation, to group-theoretic architectures with strict equivariance, to spectral and operator-based formulations with provable error bounds, and more recently to soft regularization strategies adapted to practical deep learning pipelines.

While these advances have laid a strong theoretical foundation, their adoption in downstream detection tasks, particularly in remote sensing, remains relatively limited. Most existing approaches have been developed for classification, segmentation, or low-level vision tasks, and often require redesigning entire network architectures to maintain rotation-equivariant representations. This makes them less suited for high-resolution aerial imagery, where small, densely packed, and arbitrarily oriented objects pose unique challenges.

In contrast, our work introduces a parameter-efficient modular adaptor, RAMMC, that can be seamlessly integrated into existing detection networks to explicitly encode geometric priors. To further enhance geometric consistency, we design two regularization terms and provide an implementation-aware equivariance-defect analysis for the RAMMC feature representation under stated assumptions, enabling stable training and improved rotation robustness.

Rather than modifying full architectures or focusing on isolated components such as backbones or prediction heads, our approach provides a scalable and theoretically grounded solution that bridges inductive bias with trainable regularization, making it particularly well suited for robust and adaptable object detection in complex remote sensing scenarios. Empirically, it consistently improves detection accuracy across multiple architectures and datasets, confirming its practical utility.

III. Our Method

A. Notation and Preliminary Acknowledgment

We begin by introducing the notations and geometric preliminaries used throughout this work.

1) Notation: Scalars are denoted by lowercase letters (x, α, λ), vectors by bold lowercase letters $(\mathbf{x})$, and matrices by uppercase letters (M, K, Z). Feature maps such as $F \in \mathbb{R}^{C\times H\times W}$ are treated as tensors indexed by channel and spatial coordinates.

The symbol "$*$" denotes convolution, "$\odot$" denotes the Hadamard (elementwise) product, and "$\cdot$" denotes scalar multiplication or weighting. Global average pooling is written as $\mathrm{GAP}(\cdot)$, and $\mathrm{MLP}(\cdot)$ denotes a two-layer perceptron.

We use Π_α^I to denote the geometric rotation operator acting on the input space, where α is a group element in the discrete rotation set. The corresponding induced action on the feature space is denoted by Π_α^F. When the feature tensor carries explicit orientation channels, we use ρ_α to denote the cyclic reindexing operator acting only on the orientation-channel axis. In this case, Π_α^F represents the full transformed action on the feature tensor, while ρ_α refers specifically to its orientation-channel permutation component. Group pooling, written as $\mathrm{Pool}_r(\cdot)$, averages feature responses over orientation channels to produce a rotation-invariant representation.

2) Rotation Group and Equivariance: Let $\mathcal{P}_t = \{0, 2\pi/t, \ldots, (2\pi(t-1))/t\}$ denote the discrete rotation

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group of order t. For each $\alpha \in \mathcal{P}_t$, the corresponding planar rotation matrix is

$$R_\alpha = \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}. \tag{1}$$

We use Π_α^I and Π_α^F to denote the actions of $\alpha \in \mathcal{P}_t$ on the input and feature spaces, respectively.

A mapping Φ is said to be *rotation-equivariant* if

$$\Phi\left(\Pi_\alpha^I(x)\right) = \Pi_\alpha^F(\Phi(x)) \quad \forall \alpha \in \mathcal{P}_t \tag{2}$$

while a classifier c is *rotation-invariant* if

$$c\left(\Pi_\alpha^I(x)\right) = c(x) \quad \forall \alpha \in \mathcal{P}_t. \tag{3}$$

3) Quarter-Turn Group: In this work, we mainly instantiate the above formulation with the quarter-turn group $\mathcal{P}_4 = \{0, \pi/2, \pi, 3\pi/2\}$, consistent with common conventions in discrete rotation-equivariant networks [32], [35]. This choice should be viewed as a canonical special case of the general discrete rotation group $\mathcal{P}_t$ introduced above.

4) Group Action on Orientation Channels: For a feature tensor $F = [F_\theta]_{\theta \in \mathcal{P}_4}$ with explicit orientation channels indexed by angles in $\mathcal{P}_4$, the induced feature-space action consists of two coupled parts: a spatial rotation applied to each channel slice and an orientation-channel re-indexing across the discrete angle axis. Formally

$$\Pi_\alpha^F(F) = \left[\Pi_\alpha^I\left(F_{\alpha^{-1}\circ 0}\right), \Pi_\alpha^I\left(F_{\alpha^{-1}\circ \frac{\pi}{2}}\right) \right. \\ \left. \Pi_\alpha^I\left(F_{\alpha^{-1}\circ \pi}\right), \Pi_\alpha^I\left(F_{\alpha^{-1}\circ \frac{3\pi}{2}}\right)\right] \tag{4}$$

where $\alpha^{-1}\circ\theta$ denotes the inverse group action used to re-index the orientation channels before spatial rotation. The operator ρ_α denotes only the cyclic permutation along the orientation-channel axis, whereas Π_α^F denotes the full induced action on the feature tensor.

Accordingly, group pooling induces invariance by averaging over the orientation channels

$$\mathrm{Pool}_r(F) = \frac{1}{t}\sum_{\alpha \in \mathcal{P}_t} \rho_\alpha(F). \tag{5}$$

B. Rotation-Aware Adaptor and Consistency Regularization

We now present the main methodological part of the proposed approach. The method consists of two tightly coupled components: the RAMMC adaptor for rotation-aware feature construction and a dual-level consistency regularization strategy for stabilizing feature and prediction behaviors under geometric transformations.

1) RAMMC Module: As illustrated in Fig. 3, the proposed RAMMC module is designed to construct an orientation-aware intermediate representation and its invariant counterpart in a structured manner. The module consists of two tightly coupled components. First, rotated mother kernels are constructed by generating multiple orientation-specific child kernels from each learnable mother kernel, thereby forming a compact kernel bank that explicitly encodes rotational patterns. This design introduces structured parameter sharing across discrete orientations, allowing the module to enhance directional sensitivity without linearly increasing the number of independently learned kernels. To reduce computational cost and decouple orientation-sensitive channels, the input features are first projected through a 1×1 convolution and then convolved with the rotated kernel bank.

Second, an orientation-aware aggregation mechanism is applied. Responses across orientations are modulated by both spatial attention maps and global orientation weights, and are subsequently fused to produce the orientation-explicit representation Z_{orient}. Here, $\odot$ denotes the attention-weighted feature modulation operator, while $\oplus$ indicates feature concatenation in the aggregation process. This aggregation enables the adaptor to selectively emphasize informative rotational responses while suppressing less relevant ones.

Importantly, RAMMC does not stop at producing an orientation-sensitive tensor. From Z_{orient}, we further derive its pooled counterpart

$$Z_{\text{pool}} := \mathrm{Pool}_r(Z_{\text{orient}})$$

which serves as a rotation-invariant summary of the orientation-aware response. Therefore, the module yields two complementary representations: the orientation-explicit feature Z_{orient}, which preserves directional and geometric information, and the pooled invariant feature Z_{pool}, which provides a more stable semantic descriptor under in-plane rotations.

This separation follows the distinct transformation requirements of the two detection subtasks. Classification seeks semantic consistency under input rotation and therefore benefits from an invariant representation, whereas oriented box regression requires sensitivity to geometric orientation and is more naturally supported by an equivariant or orientation-explicit representation. In this sense, Z_{pool} is better aligned with the needs of classification, while Z_{orient} is the appropriate carrier of directional information for localization and angle prediction.

To further improve geometric stability, feature-level and prediction-level regularizers are incorporated to constrain the intermediate orientation-explicit representation and the final detection behavior, respectively. Combined with the group-structured construction of RAMMC, this design establishes a coherent path from equivariant feature generation to invariant semantic summarization. The corresponding analytical support is given in Section III-D.

a) Rotated Mother Kernels: Let $X \in \mathbb{R}^{C_{\text{in}} \times H \times W}$ denote the input feature map, where C_{in} is the number of input channels. We define M learnable *mother kernels* $\{K_m\}_{m=1}^M$, each with shape $\mathbb{R}^{C_{\text{in}} \times k \times k}$. For each mother kernel K_m, we generate its rotated versions over a discrete rotation set $\mathcal{P}_t = \{0, 2\pi/t, \ldots, (2\pi(t-1))/t\}$

$$K_{m,\alpha}(\mathbf{u}) = K_m(R_{-\alpha}\mathbf{u}), \quad \alpha \in \mathcal{P}_t, \ \mathbf{u} \in \mathbb{Z}^2.$$

This yields a bank of $M \times t$ orientation-specific kernels. The number of mother kernels M is a tunable hyperparameter that controls the capacity of the rotation branch and is independent of the rotation order t. By construction, this kernel bank couples multiple orientations through a shared mother-kernel parameterization, which improves rotational coverage while keeping the branch parameter-efficient.

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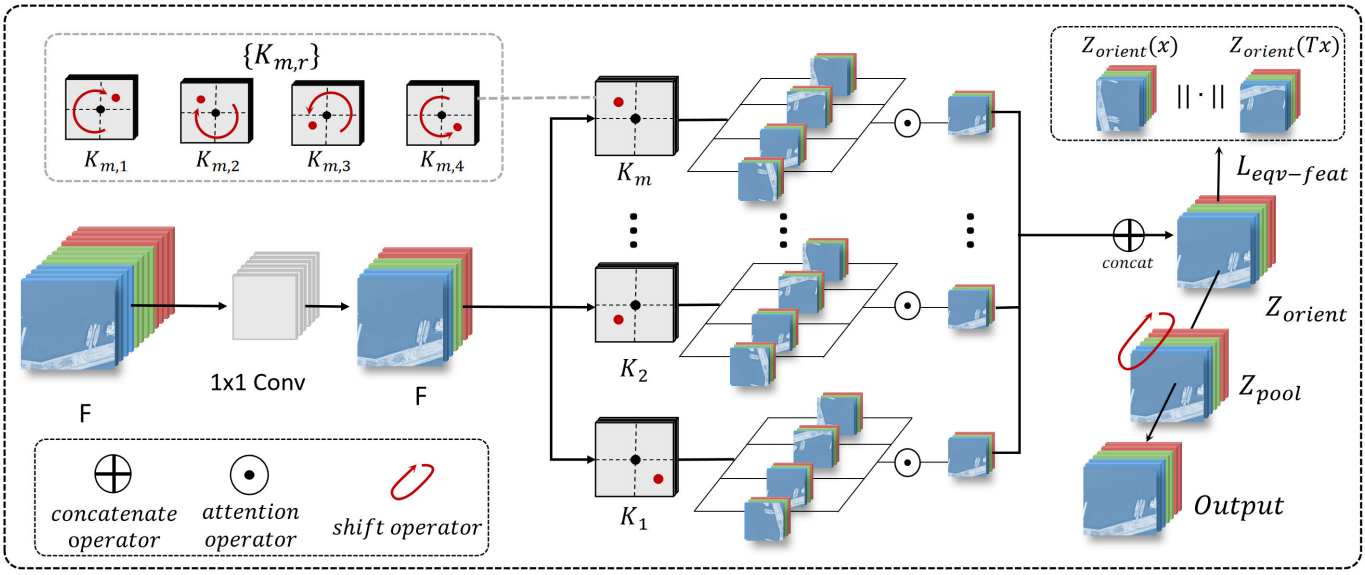


Fig. 3. Illustration of the proposed RAMMC module. The input feature F is first projected by a 1×1 convolution and then processed by rotated mother kernels $\{K_{m,r}\}$. The resulting branch features are modulated by orientation attention and concatenated to form the orientation-explicit representation Z_{orient}. The feature-level equivariance regularization term $\mathcal{L}_{\text{eqv-feat}}$ is imposed on Z_{orient} by comparing the feature of the original sample with the aligned feature of its transformed counterpart. After the orientation channel shift and pooling operation, a pooled representation is obtained and further used to produce the module output. In the diagram, $\odot$ denotes the attention-weighted feature modulation operator, while $\oplus$ denotes feature concatenation.

b) Rotation-Aware Aggregation: To reduce computational cost, the input feature X is first projected to a lower-dimensional rotation-branch representation $X_{\text{rot}} \in \mathbb{R}^{C_{\text{rot}} \times H \times W}$ via a 1×1 convolution, where $C_{\text{rot}} < C_{\text{in}}$ controls the channel budget of the rotation branch. For each mother kernel K_m, we compute t oriented convolutions $\{Y_{m,\alpha}\}$ and aggregate them through a two-branch mechanism

$$Z_m = \sum_{\alpha \in \mathcal{P}_t} A_{m,\alpha} \cdot \left(S_{m,\alpha} \odot Y_{m,\alpha}\right)$$

where $S_{m,\alpha}$ denotes a spatial attention map obtained from a compact squeeze-and-excitation (SE) or selective-kernel (SK) module, and $A_{m,\alpha}$ is a global orientation weight produced by a softmax-based attention MLP. The outputs of all M branches are then concatenated to form

$$Z_{\text{orient}} = \text{Concat}\left(Z_1, \ldots, Z_M\right)$$

which is used as the orientation-enhanced feature representation in the subsequent fusion stage. This aggregation combines local spatial saliency with global orientation preference, producing an orientation-explicit feature that preserves directional cues. Its pooled counterpart provides a more invariant descriptor, while Z_{orient} itself retains the geometric sensitivity required for subsequent regression.

2) Dual-Level Consistency Regularization: Although RAMMC enhances rotation-aware feature construction, explicit orientation modeling alone does not guarantee consistent detector behavior under input rotation. In practice, inconsistency may arise both in the intermediate representation and in the final decoded detections. We therefore impose consistency regularization at both the feature and prediction levels.

a) Feature-Level Regularizer: The first term acts on the intermediate orientation-explicit representation produced by RAMMC

$$\mathcal{L}_{\text{eqv-feat}} = \mathbb{E}_{\alpha \sim \mathcal{P}_t} \left\| Z_{\text{orient}}^{(\alpha)} - \Pi_\alpha^F \left(Z_{\text{orient}}^{(0)}\right) \right\|_2 \tag{6}$$

where $Z_{\text{orient}}^{(\alpha)}$ and $Z_{\text{orient}}^{(0)}$ denote the orientation-explicit RAMMC outputs for the rotated and original inputs, respectively, and Π_α^F is the induced feature-space action defined in (4). This regularizer explicitly requires that the orientation-aware feature extracted from a rotated input remain consistent with the transformed counterpart of the original feature under the induced group action.

This choice is mathematically aligned with the role of Z_{orient}. Since Z_{orient} is the representation expected to preserve directional and geometric information, it is also the natural object on which approximate equivariance should be enforced. By minimizing the deviation between $Z_{\text{orient}}^{(\alpha)}$ and $\Pi_\alpha^F(Z_{\text{orient}}^{(0)})$, the regularizer directly suppresses rotation-induced drift in the intermediate geometry-aware representation, thereby stabilizing the subsequent regression pathway and improving the consistency of feature formation across transformed inputs.

b) Prediction-Level Regularizer: To further constrain the detector at the output level, we introduce a prediction-level consistency regularizer that compares the decoded detections of the original input with those of the rotated input after mapping them back to the same coordinate system. Let

$$\mathcal{B}^{(0)} = \mathcal{Q}\left(f\left(x\right)\right)$$

denote the set of decoded OBBs predicted from the original input, and let

$$\mathcal{B}^{(\alpha \to 0)} = \Pi_{-\alpha}^I \left(\mathcal{Q}\left(f\left(\Pi_\alpha^I x\right)\right)\right)$$

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denote the prediction set obtained from the rotated input and transformed back to the canonical coordinate system. Since the two prediction sets are unordered and generally have no one-to-one correspondence a priori, their comparison is formulated as a differentiable soft matching problem.

Specifically, for each rotation $\alpha \in \mathcal{P}_t$, we define a pairwise discrepancy matrix

$$C_{ij}^{(\alpha)} = \delta\left(b_i^{(0)}, b_j^{(\alpha\to 0)}\right), \quad b_i^{(0)} \in \mathcal{B}^{(0)},\ b_j^{(\alpha\to 0)} \in \mathcal{B}^{(\alpha\to 0)}$$

where $\delta(\cdot,\cdot)$ is a differentiable discrepancy between two oriented boxes. A typical choice is

$$\delta\left(b_i, b_j\right) = \lambda_{\text{iou}}\left(1 - \text{IoU}_{\text{soft}}\left(b_i, b_j\right)\right) + \lambda_{\text{ctr}}\left\|\mathbf{c}_i - \mathbf{c}_j\right\|_1 + \lambda_{\text{size}}\left\|\mathbf{s}_i - \mathbf{s}_j\right\|_1 + \lambda_{\text{ang}}\, d_{\text{ang}}\left(\theta_i, \theta_j\right)$$

where $\mathbf{c}$, $\mathbf{s}$, and θ denote the box center, size, and orientation, respectively, and IoU_{soft} denotes a differentiable IoU surrogate.

To obtain a differentiable correspondence between the two prediction sets, we replace hard assignment by an entropy-regularized transport plan $P^{(\alpha)}$, computed via Sinkhorn iterations. The prediction-level regularizer is then defined as follows:

$$\mathcal{L}_{\text{eqv-pred}} = \mathbb{E}_{\alpha\sim\mathcal{P}_t}\left[\left\langle P^{(\alpha)}, C^{(\alpha)}\right\rangle\right] \tag{7}$$

where $\langle\cdot,\cdot\rangle$ denotes the Frobenius inner product. This term directly penalizes inconsistency in the final detection behavior, including localization, scale, and orientation, after aligning the predictions to a shared reference frame.

c) Complementary effect: The two regularizers play complementary roles: the feature-level term stabilizes the formation of orientation-aware representations, while the prediction-level term constrains how these representations are translated into final geometric outputs. Their combination therefore provides a more complete consistency constraint across the detection pipeline than either term alone.

d) Unified loss: The final training objective includes both regularizers

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{det}} + \lambda_{\text{feat}}\mathcal{L}_{\text{eqv-feat}} + \lambda_{\text{pred}}\mathcal{L}_{\text{eqv-pred}} \tag{8}$$

where $\mathcal{L}_{\text{det}}$ is the standard detection loss, and $\lambda_{\text{feat}}, \lambda_{\text{pred}}$ are balancing weights. Under the adopted differentiable IoU surrogate and soft assignment formulation, both regularizers are differentiable with respect to the detector outputs and can therefore be optimized jointly with the standard detection objective through backpropagation. As a result, the proposed loss not only preserves the original detection objective, but also provides an explicit and trainable constraint toward rotation-consistent representations and predictions.

e) Analytical Support: From an analytical perspective, the two regularizers act on complementary stages of the detection pipeline and jointly reduce the equivariance defect induced by input rotation. Their formal characterization is provided in Section III-D.

C. Algorithm Instantiation

We instantiate the proposed RAMMC adaptor on a YOLO-style detector, as illustrated in Fig. 2. The backbone's stride-{8, 16} feature maps are augmented with an orientation-aware path, while the neck and head remain unchanged.

1) Backbone Integration: Given a feature map $F^{(s)} \in \mathbb{R}^{C\times H\times W}$ at stride s, we first apply a 1×1 projection to obtain the rotation-branch input $X_{\text{rot}} \in \mathbb{R}^{C_{\text{rot}}\times H\times W}$. The RAMMC adaptor then produces an orientation-explicit representation

$$Z_{\text{orient}} = \text{RAMMC}\left(X_{\text{rot}}\right)$$

which preserves directional and geometric information through rotated mother kernels and rotation-aware aggregation. Its invariant counterpart is subsequently obtained by group pooling

$$Z_{\text{pool}} = \text{Pool}_r\left(Z_{\text{orient}}\right).$$

Accordingly, the instantiated detector maintains two complementary streams at each selected backbone level: an orientation-explicit stream Z_{orient} for geometry-sensitive prediction, and an invariant stream Z_{pool} for rotation-robust semantic encoding.

2) Stream Fusion: The invariant stream is injected back into the backbone pathway through a channel gate

$$Y_{\text{fused}} = (1-g)\odot F^{(s)} + g\odot Z_{\text{pool}} \tag{9}$$

where g is derived from a global context descriptor obtained via global average pooling followed by a two-layer MLP. Here, Y_{fused} should be understood as an invariant-enhanced semantic feature, rather than a replacement for the orientation-explicit branch. In the resulting design, the classification branch primarily relies on Y_{fused}, since category prediction ideally remains stable under in-plane rotation, whereas the regression branch operates on Z_{orient} to preserve directional cues for localization and angle estimation.

This routing follows the transformation requirements of the two subtasks: classification favors invariant semantics, whereas oriented regression requires directional sensitivity. Accordingly, the fusion mechanism injects invariant semantic information into the shared feature stream while preserving the orientation-explicit representation for geometry-aware prediction.

3) Training Objective: Training adopts the objective in (8), where the standard detection loss is augmented by the two consistency regularizers

$$\mathcal{L} = \mathcal{L}_{\text{det}} + \lambda_{\text{feat}}\mathcal{L}_{\text{eqv-feat}} + \lambda_{\text{pred}}\mathcal{L}_{\text{eqv-pred}} \tag{10}$$

with $\mathcal{L}_{\text{eqv-feat}}$ and $\mathcal{L}_{\text{eqv-pred}}$ defined in Section III-B. The prediction-level regularizer in (7) compares two unordered sets of decoded detections through a soft matching matrix and a differentiable pairwise box discrepancy. To enable backpropagation through this objective, we adopt two differentiable relaxations:

1) *Differentiable Geometric Discrepancy.* The pairwise box discrepancy $\delta(\cdot,\cdot)$ is built upon differentiable IoU surrogates such as GIoU, DIoU, or CIoU [16], [55], optionally combined with smooth penalties on center, size, and angle parameters. This ensures that the prediction consistency term provides informative gradients even when two boxes only partially overlap.
2) *Soft Assignment.* Instead of hard one-to-one matching, which is nondifferentiable, we adopt entropy-regularized optimal transport solved via Sinkhorn iterations [56],

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[58]. The resulting doubly-stochastic transport matrix serves as a smooth approximation to discrete assignment and enables end-to-end optimization of the set-level prediction consistency objective.

This instantiation illustrates how RAMMC can be integrated into a detection pipeline through task-aligned feature routing. In the experimental section, we further evaluate its transferability across different detector architectures.

D. Implementation-Aware Theoretical Analysis

This section provides an implementation-aware analysis of the design motivation discussed in Section III-B. Instead of claiming unconditional exact equivariance for the whole detector, we analyze the conditional equivariance defect induced by the implemented RAMMC branch and clarify how its main constants can be computed or monitored from the actual model parameters.

Definition 1 (Equivariance defect): Let Φ be a detection network and x an input image. For a discrete rotation $\alpha \in \mathcal{P}_t$, the equivariance defect is defined as follows:

$$\mathcal{E}_\alpha(x) := \left\| \Phi\left(\Pi_\alpha^I x\right) - \Pi_\alpha^F \Phi(x) \right\| \tag{11}$$

which measures the deviation from perfect equivariance under the induced feature-space action. When Φ is instantiated by RAMMC, this quantity characterizes the mismatch between the actual orientation-aware representation and its ideal transformed counterpart under input rotation.

Lemma 1 (Reynolds Operator Induced by Group Pooling): Let $G = \mathcal{P}_t$ be the discrete rotation group, and let ρ_g denote the induced action of $g \in G$ on the orientation-channel axis of an orientation-explicit feature tensor Z. The Reynolds operator associated with G is defined as follows:

$$\mathcal{R}_G Z := \frac{1}{|G|} \sum_{g \in G} \rho_g Z. \tag{12}$$

For the RAMMC representation $Z_{\text{orient}} \in \mathbb{R}^{(tC)\times H\times W}$, the orientation group-pooling operation is exactly this Reynolds averaging operator

$$Z_{\text{pool}} := \text{Pool}_r(Z_{\text{orient}}) = \mathcal{R}_{\mathcal{P}_t} Z_{\text{orient}} = \frac{1}{|\mathcal{P}_t|} \sum_{\alpha \in \mathcal{P}_t} \rho_\alpha Z_{\text{orient}}. \tag{13}$$

The operator $\mathcal{R}_G$ is invariant and idempotent, namely

$$\begin{aligned} \rho_h \mathcal{R}_G Z &= \mathcal{R}_G Z \\ \mathcal{R}_G \rho_h Z &= \mathcal{R}_G Z \\ \mathcal{R}_G^2 Z &= \mathcal{R}_G Z \quad \forall h \in G. \end{aligned} \tag{14}$$

Consequently, Z_{pool} is invariant under the discrete rotation action.

Proof: For any $h \in G$, using the representation property $\rho_h \rho_g = \rho_{h\circ g}$, we have

$$\begin{aligned} \rho_h \mathcal{R}_G Z &= \frac{1}{|G|} \sum_{g \in G} \rho_h \rho_g Z \\ &= \frac{1}{|G|} \sum_{g \in G} \rho_{h\circ g} Z. \end{aligned} \tag{15}$$

Since left multiplication by h only permutes the elements of G, $\{h \circ g : g \in G\} = G$, and thus $\rho_h \mathcal{R}_G Z = \mathcal{R}_G Z$. Similarly, right multiplication by h permutes G, which gives $\mathcal{R}_G \rho_h Z = \mathcal{R}_G Z$. Finally

$$\begin{aligned} \mathcal{R}_G^2 Z &= \frac{1}{|G|^2} \sum_{h \in G} \sum_{g \in G} \rho_h \rho_g Z \\ &= \frac{1}{|G|^2} \sum_{h,g \in G} \rho_{h\circ g} Z. \end{aligned} \tag{16}$$

For each $u \in G$, there are exactly $|G|$ pairs (h, g) such that $h \circ g = u$. Therefore,

$$\mathcal{R}_G^2 Z = \frac{1}{|G|} \sum_{u \in G} \rho_u Z = \mathcal{R}_G Z. \tag{17}$$

Taking $G = \mathcal{P}_t$ and $Z = Z_{\text{orient}}$ proves that $Z_{\text{pool}} = \mathcal{R}_{\mathcal{P}_t} Z_{\text{orient}}$ is invariant under the discrete rotation action. □

Proposition 1 (Ideal equivariance of the RAMMC orientation-explicit representation): Let X_{rot} denote the input feature to the RAMMC branch, and let

$$Z_{\text{orient}} = \text{RAMMC}(X_{\text{rot}})$$

be the resulting orientation-explicit representation. Under the rotated-kernel construction and the group-consistent tying of spatial gating maps and orientation weights across discrete orientations, the RAMMC mapping is equivariant with respect to the group action, i.e.,

$$\text{RAMMC}\left(\Pi_\alpha^I X_{\text{rot}}\right) = \Pi_\alpha^F\left(\text{RAMMC}(X_{\text{rot}})\right) \quad \forall \alpha \in \mathcal{P}_t. \tag{18}$$

Proof sketch: The 1×1 projection commutes with spatial rotation because it acts only along the channel dimension. The rotated mother-kernel bank is constructed by index-consistent transformations of shared kernels, so the convolutional responses inherit the corresponding rotated-shifted structure. Under the group-consistent tying condition, both the spatial gating maps and the orientation weights transform coherently with the rotated input. Therefore, the aggregated orientation responses remain aligned with the induced feature-space action. Combining these steps yields

$$\text{RAMMC}\left(\Pi_\alpha^I X_{\text{rot}}\right) = \Pi_\alpha^F\left(\text{RAMMC}(X_{\text{rot}})\right)$$

which proves the equivariance of the orientation-explicit output Z_{orient}. □

1) Implementation-Aware Lipschitz Bounds: We next make the constants in the nonideal case explicit. Let the spatial gating module and the orientation-attention module be written as follows:

$$S(Y) = \sigma\left(W_2^S \phi\left(W_1^S C_S(Y)\right)\right) \tag{19}$$

$$A(Y) = \text{softmax}\left(\frac{W_2^A \phi\left(W_1^A C_A(Y)\right)}{\tau}\right) \tag{20}$$

where Y denotes the orientation-indexed preaggregation response, C_S and C_A are the context operators, $W_1^S, W_2^S, W_1^A, W_2^A$ are the implemented projection layers, ϕ is a 1-Lipschitz activation, σ is the sigmoid function, and $\tau > 0$ is the softmax temperature. Since the sigmoid is 1/4-Lipschitz, the temperature-scaled softmax is at most $1/(2\tau)$-Lipschitz under the Euclidean norm, and the Lipschitz constant of a linear layer equals its spectral norm, we obtain

$$L_S = \text{Lip}(S) \le \frac{1}{4} \left\|W_2^S\right\|_2 \left\|W_1^S\right\|_2 \text{Lip}(C_S) \tag{21}$$

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$$L_A = \mathrm{Lip}(A) \le \tfrac{1}{2\tau} \left\|W_2^A\right\|_2 \left\|W_1^A\right\|_2 \mathrm{Lip}(C_A). \tag{22}$$

When C_S or C_A is implemented by global average pooling over an $H \times W$ feature map, its Frobenius-to-Euclidean Lipschitz constant is bounded by $1/(HW)^{1/2}$ and can be conservatively upper-bounded by 1. Therefore, the constants used in the analysis are not abstract assumptions. In practice, they can be estimated as follows:

$$\begin{aligned} \widehat{L}_S &= \tfrac{1}{4} \widehat{\sigma}\left(W_2^S\right) \widehat{\sigma}\left(W_1^S\right) \mathrm{Lip}(C_S) \\ \widehat{L}_A &= \tfrac{1}{2\tau} \widehat{\sigma}\left(W_2^A\right) \widehat{\sigma}\left(W_1^A\right) \mathrm{Lip}(C_A) \end{aligned} \tag{23}$$

where $\widehat{\sigma}(\cdot)$ denotes a spectral-norm estimate obtained by power iteration. These estimates can be monitored during training or controlled by weight decay, spectral regularization, or the temperature τ.

Theorem 1 (Computable Equivariance-Defect Decomposition Under RAMMC): Let $Y(X)$ denote the rotated mother-kernel response before gating and attention aggregation. Assume that the finite rotated-kernel implementation satisfies

$$\left\|Y\left(\Pi_\alpha^I X\right) - \Pi_\alpha^F Y(X)\right\| \le C_1 h^2 + C_2 \tfrac{ph}{t} \tag{24}$$

where h is the grid spacing, p is the kernel radius, and t is the number of discrete orientations. Assume further that the preaggregation response is bounded by B_Y on the feature region encountered during training. Define the context mismatch terms

$$\Delta_\alpha^S = C_S\left(Y\left(\Pi_\alpha^I X\right)\right) - \Pi_\alpha^S C_S(Y(X)) \tag{25}$$

$$\Delta_\alpha^A = C_A\left(Y\left(\Pi_\alpha^I X\right)\right) - \rho_\alpha C_A(Y(X)). \tag{26}$$

Then the equivariance defect of the orientation-explicit RAMMC representation satisfies

$$\begin{aligned} &\left\|Z_{\text{orient}}\left(\Pi_\alpha^I X\right) - \Pi_\alpha^F Z_{\text{orient}}(X)\right\| \\ &\le C_K\left(C_1 h^2 + C_2 \tfrac{ph}{t}\right) \\ &\quad + C_S' B_Y L_S \left\|\Delta_\alpha^S\right\| \\ &\quad + C_A' B_Y L_A \left\|\Delta_\alpha^A\right\| \end{aligned} \tag{27}$$

where C_K, C_S', and C_A' absorb constants related to the number of branches, orientations, and broadcasting operations.

Proof sketch: The ideal rotated mother-kernel bank is equivariant by Proposition 1. In the discrete implementation, interpolation and finite orientation sampling introduce the residual in (24). The attention-modulated RAMMC response can be written branch-wise as follows:

$$\widetilde{Z}_{m,k}(X) = A_{m,k}(Y(X))\left(S_{m,k}(Y(X)) \odot Y_{m,k}(X)\right). \tag{28}$$

Subtracting the aligned response under Π_α^F and applying the triangle inequality gives three terms: the preaggregation kernel residual, the spatial-gating mismatch, and the orientation-attention mismatch. Since sigmoid gates and softmax weights are bounded in $[0,1]$ and $\|Y\| \le B_Y$, the latter two terms are amplified by B_Y. Finally, applying (21) and (22) to the context mismatches in (25) and (26) yields (27). □

Proposition 2 (Dual-level consistency regularization controls the equivariance defect): Let the RAMMC-enhanced detector be trained with the two regularizers $\mathcal{L}_{\text{eqv-feat}}$ and $\mathcal{L}_{\text{eqv-pred}}$, where the former acts on the orientation-explicit intermediate representation Z_{orient} and the latter acts on the decoded detection outputs. Under a Lipschitz-continuous prediction head and a differentiable matching discrepancy, the expected equivariance defect satisfies

$$\mathbb{E}_\alpha\, \mathcal{E}_\alpha(x) \le \lambda_1\, \mathcal{L}_{\text{eqv-feat}} + \lambda_2\, \mathcal{L}_{\text{eqv-pred}} + \bar{\epsilon} \tag{29}$$

where $\bar{\epsilon}$ collects approximation residuals induced by finite kernel rotation, interpolation, and attention projection.

Interpretation. The feature-level regularizer suppresses deviation from the ideal equivariant trajectory at the representation level, while the prediction-level regularizer constrains the residual mismatch after decoding. Together, they provide a soft control on the end-to-end equivariance defect.

2) Practical Implication: Equation (27) separates the RAMMC equivariance defect into three implementation-relevant sources: the finite kernel-rotation residual, the spatial-gating context mismatch scaled by L_S, and the orientation-attention context mismatch scaled by L_A. Because L_S and L_A are explicit functions of the spectral norms of the implemented projection layers and the softmax temperature, the bound provides direct guidance for model design and training. Increasing the orientation resolution t reduces the angular discretization component, controlling the spectral norms of the gating and attention projections limits the amplification of context mismatch, and increasing τ improves the stability of orientation aggregation at the cost of less sharply peaked attention.

3) Discussion: The above results provide a coherent but conditional explanation of the proposed design. Proposition 1 shows the ideal equivariant behavior targeted by the RAMMC construction, Lemma 1 explains why group pooling produces an invariant semantic stream, and Theorem 1 identifies the practical residuals that remain in the implemented network. Proposition 2 further explains why the dual-level regularization is useful in practice: it suppresses equivariance error both before and after prediction, thereby reducing the mismatch between geometric transformations in the input space and their induced responses in the detector.

IV. Experiments

We conduct comprehensive experiments to demonstrate the effectiveness and versatility of the proposed framework. Built upon YOLO-based oriented object detection, our method achieves notable accuracy improvements, and we further validate its adaptability by integrating it into other detection backbones. Section IV-A detail the benchmark datasets, implementation settings, comparisons with state-of-the-art methods, and ablation studies.

A. Datasets

We evaluate our method on three authoritative benchmarks for oriented object detection in remote sensing imagery.

1) *DOTA v1.0* [8]: One of the largest and most challenging benchmarks, containing 2806 aerial images (800 × 800 to 4000 × 4000) with over 188 000 OBBs across 15 categories, such as plane, ship, storage tank, bridge,

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large vehicle, small vehicle, harbor, and helicopter. The dataset is split into 1411 training, 937 validation, and 458 test images, and is evaluated by mAP at IoU threshold 0.5.

2) *FAIR1M v1* [25]: A fine-grained dataset with 15 000 high-resolution images (up to 8000 × 8000) and over 1 million object instances annotated across 37 categories. It emphasizes aircraft, ships, and vehicles, with splits of 6:2:2 for training, validation, and test. Compared with DOTA, FAIR1M v1 is more challenging in recognizing fine-grained subcategories (e.g., Boeing versus Airbus).
3) *FAIR1M v2*: An extended version with over two million instances across the same 37 categories. It provides greater diversity, more balanced categories, and improved annotation quality, making it a realistic benchmark for dense and complex scenes.

B. Implementation Details

We implement the proposed RAMMC module on top of Ultralytics YOLO [59], [65], [66] and MMRotate [21], covering both anchor-free and anchor-based detectors. For YOLO, we follow the default augmentation pipeline including random resizing, HSV/color jittering, Mosaic, and MixUp. For MMRotate, training is conducted on full images after multiscale resizing (*no cropping or tiling*), together with random rotation, flipping, and photometric distortions.

1) Module Instantiation: In each backbone block, the input feature is first projected by a 1×1 convolution with reduction ratio $\beta = 0.5$, yielding the rotated-path feature. RAMMC is then instantiated with $M = 4$ mother kernels, each expanded into $K = 4$ rotated replicas (uniformly sampled over $[0, 2\pi)$). Low-cost gating and attention mechanisms are employed, including SK gating, orientation attention across rotated replicas, and a channel gate for fusing rotated and vanilla streams. The orientation-consistency regularizer is applied with weight $\lambda = 0.1$ by default.

2) Differentiable Approximations: For the prediction-level regularizer, we adopt *CIoU* [16] as the surrogate to standard IoU, since it provides stable gradients even when boxes do not overlap. For the assignment step, we use *Sinkhorn-based optimal transport* [56] with entropy regularization (five iterations, $\epsilon = 0.05$), which approximates Hungarian matching by a differentiable doubly-stochastic matrix. These settings are fixed throughout all experiments.

3) Optimization: All models are trained on 8× NVIDIA RTX 3090 GPUs with a batch size of 16 per GPU, using SGD with momentum 0.9, weight decay 10^{-4}, and a cosine learning-rate schedule. Automatic mixed-precision (AMP) is enabled for efficiency.

4) Inference Protocol: We evaluate under both *nonslicing* (whole-image) inference and *slicing* inference (overlapping 1024×1024 tiles with merged NMS) to ensure fair comparison with prior works and to reflect practical deployment.

C. Evaluation Metrics

To comprehensively evaluate the proposed method, we adopt standard metrics widely used in oriented object detection and model efficiency analysis.

1) Average Precision at IoU 0.5 (AP_{50}): Following the VOC convention [57], we report AP at IoU threshold 0.5. For a given class c, let $\mathcal{P}_c(r)$ denote the precision as a function of recall $r \in [0, 1]$

$$\mathcal{P}_c(r) = \frac{TP_c(r)}{TP_c(r) + FP_c(r)}$$
$$\mathcal{R}_c(r) = \frac{TP_c(r)}{TP_c(r) + FN_c(r)} \tag{30}$$

where TP_c, FP_c, and FN_c denote true positives, false positives, and false negatives. The class-wise AP is the area under the precision–recall curve

$$AP_{50}(c) = \int_0^1 \mathcal{P}_c(r)\, dr \tag{31}$$

and the final score is averaged over C classes

$$AP_{50} = \frac{1}{C} \sum_{c=1}^{C} AP_{50}(c). \tag{32}$$

2) Intersection Over Union: For a predicted oriented bounding box B_p and ground-truth B_{gt}

$$\text{IoU}\left(B_p, B_{gt}\right) = \frac{\left|B_p \cap B_{gt}\right|}{\left|B_p \cup B_{gt}\right|}. \tag{33}$$

3) Confusion Matrix: We compute the confusion matrix $M \in \mathbb{N}^{C \times C}$ with entries

$$M_{i,j} = \#\left\{ x \mid y_{\text{gt}}(x) = i,\ y_{\text{pred}}(x) = j \right\} \tag{34}$$

where y_{gt} and y_{pred} denote ground-truth and predicted labels. Diagonal entries $M_{i,i}$ represent correct predictions, off-diagonal entries correspond to misclassifications.

4) Model Size (Parameters): The number of learnable parameters is

$$\text{Params} = \sum_{l=1}^{L} |\theta_l| \tag{35}$$

where $|\theta_l|$ denotes the number of scalar entries in the parameter tensor of layer l.

5) Inference Time: The average per-image inference latency is

$$\text{Latency} = \frac{1}{N} \sum_{i=1}^{N} T_i \tag{36}$$

where T_i is the processing time for the ith test image.

These metrics jointly evaluate both detection accuracy, including AP_{50}, IoU, and the confusion matrix, and computational efficiency in terms of the number of parameters and latency.

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TABLE I

PER-CATEGORY AP_{50} COMPARISON ON THE FAIR1M-V2.0 DATASET. OUR METHOD ACHIEVES THE BEST RESULTS ACROSS MOST FINE-GRAINED CATEGORIES

Category	Sub Category	RetinaNet [61]	GWD [22]	KLD [64]	S2ANet [30]	R3Det [11]	FCOS [62]	FRCNN [60]	GV [10]	RoI Trans [9]	ReDet [15]	ORCNN [12]	PETDet [72]	YOLO8-n [59]	YOLO11-n [65]	DRNet+ [63]	**Ours**
Airplane	B737	37.60	38.05	38.82	39.74	39.28	38.12	42.29	41.36	46.52	44.02	44.16	49.96	13.60	17.60	45.20	50.40
Airplane	B747	86.54	87.59	88.53	88.74	87.47	89.30	92.73	93.02	93.05	94.41	93.13	93.60	56.90	65.60	92.35	95.80
Airplane	B777	28.21	28.66	31.68	33.51	31.33	35.65	35.68	35.67	41.60	42.89	39.64	43.77	21.40	33.70	42.63	49.20
Airplane	B787	48.91	50.52	55.97	52.09	53.93	54.72	60.17	58.94	62.88	61.41	61.64	67.27	9.15	10.50	62.65	63.70
Airplane	C919	0.44	0.35	0.35	0.62	0.70	0.80	9.06	2.13	9.90	9.29	2.25	34.53	26.30	28.20	22.89	44.00
Airplane	A220	51.19	52.37	50.12	49.34	51.77	52.85	53.51	53.51	53.80	56.03	54.35	59.91	26.00	31.10	52.00	62.00
Airplane	A321	65.89	68.87	69.13	70.52	69.87	69.63	68.34	67.03	71.07	71.81	70.61	76.72	55.00	61.90	69.16	80.00
Airplane	A330	27.86	33.79	40.78	47.54	44.99	50.31	58.69	58.21	61.48	62.98	59.94	64.33	34.90	34.10	57.91	63.90
Airplane	A350	50.82	48.41	49.39	47.07	55.16	58.98	67.20	66.14	67.02	73.09	67.54	73.29	58.80	61.90	72.56	71.20
Airplane	ARJ21	3.47	3.34	3.82	2.20	2.75	3.15	9.97	10.10	14.35	17.42	11.97	15.29	8.51	14.10	15.08	17.30
Ship	PS	1.92	4.76	4.56	1.80	9.22	12.00	12.09	13.06	12.71	15.52	15.20	13.43	10.50	8.88	14.80	14.00
Ship	MB	10.74	36.16	35.69	23.47	38.26	54.12	53.17	54.81	58.74	61.36	59.62	64.15	50.30	54.00	67.72	63.60
Ship	FB	1.43	5.51	5.80	1.55	9.55	13.93	13.93	15.15	19.33	29.59	26.46	24.94	24.10	24.70	34.15	23.50
Ship	TB	12.87	21.55	22.22	16.29	25.39	29.83	29.32	29.34	30.79	30.71	30.23	30.05	7.21	15.70	28.44	35.80
Ship	ES	4.32	12.29	11.91	3.11	10.44	13.81	20.22	13.81	20.22	15.92	17.82		4.50	0.50	*21.31*	25.10
Ship	LCS	8.33	17.98	17.35	7.19	46.48	46.47	43.29	43.76	46.96	49.78	49.06	50.67	17.30	19.20	25.30	53.50
Ship	DCS	14.11	28.86	28.96	11.96	45.48	44.45	44.45	44.08	50.52	52.59	51.12	51.18	12.50	49.40	49.56	52.40
Ship	WS	1.21	5.36	5.77	7.16	22.32	28.82	30.23	27.33	33.18	32.37	32.04		9.50	2.29	34.57	38.70
Vehicle	SC	30.40	46.80	49.61	47.77	51.17	54.92	53.78	53.32	58.10	60.00	56.87	72.00	59.60	62.50	76.09	67.10
Vehicle	BUS	2.69	5.79	7.08	3.11	2.38	11.36	11.36	3.11	31.36	23.88	30.46	43.51	10.80	9.47	49.41	54.20
Vehicle	CT	8.49	25.45	23.62	19.75	34.27	44.78	47.36	47.42	49.76	49.07	49.95	55.60	24.50	27.00	57.94	57.70
Vehicle	DT	10.09	17.67	17.67	14.13	27.98	34.71	47.07	45.94	48.78	49.52	48.82	54.76	31.70	34.30	58.91	51.10
Vehicle	VAN	20.81	38.69	41.49	29.50	43.50	51.71	47.86	47.01	52.29	52.57	51.39	69.83	57.50	60.20	74.82	60.80
Vehicle	TRI	0.00	0.12	0.00	0.26	10.43	13.12	13.68	13.12	13.68	15.99	14.24		16.10	16.80	19.27	18.00
Vehicle	TRC	0.00	0.00	0.00	0.00	0.26	1.24	1.58	1.02	1.81	1.33	1.85	1.94	3.40	0.90	2.04	12.80
Vehicle	EX	0.16	0.00	0.42	1.71	4.84	13.39	13.22	13.22	13.31	14.28	15.26	14.82	25.30	17.50	27.21	22.20
Vehicle	TT	0.00	0.18	0.03	0.02	0.42	22.79	13.45	10.22	16.64	21.20	5.39	33.14	16.30	13.90	42.30	42.70
Court	BC	30.54	38.45	38.23	40.69	47.66	60.64	55.38	54.34	55.90	64.65	57.56	54.88	64.30	67.00	60.29	77.50
Court	TC	74.59	88.53	93.23	87.14	85.68	88.63	89.55	85.83	89.55	89.55	88.56	88.17	75.40	85.10	89.93	94.60
Court	FF	54.73	59.55	61.54	57.83	59.07	57.14	57.83	60.22	62.80	66.89	63.34	67.56	60.10	60.10	63.81	70.20
Court	BF	87.28	89.61	90.31	87.58	90.25	91.83	91.47	91.74	91.74	92.36	90.90	92.36	87.50	91.00	91.55	88.10
Road	IS	60.07	62.27	60.32	56.07	61.23	65.88	65.04	63.41	64.40	65.20	64.57	65.75	55.70	68.40	63.13	67.50
Road	RA	35.13	32.63	33.21	29.05	33.71	37.13	37.13	27.51	32.84	32.84	31.24	35.73	26.10	26.80	28.23	39.90
Road	BR	15.46	25.91	23.72	18.12	25.97	32.89	32.18	31.38	32.99	39.25	35.36	32.41	27.50	32.00	34.85	48.90
mAP50		26.07	31.48	32.12	30.93	36.04	39.98	41.64	41.27	44.03	46.03	43.90	48.81	31.09	34.47	49.94	52.28

TABLE II

COMPARISON WITH A UNIFIED SET OF REPRESENTATIVE ORIENTED OBJECT DETECTION METHODS ON DOTA-V1.0

Method	Venue	Year	Backbone / Framework	Protocol	AP_{50}
Faster R-CNN [60]	NeurIPS	2015	R50-FPN	reproduced under our protocol	77.63
RetinaNet [61]	ICCV	2017	R50-FPN	reproduced under our protocol	74.09
FCOS [62]	ICCV	2019	R50-FPN	reproduced under our protocol	76.88
RoI Transformer [9]	CVPR	2019	ResNet-50	single-scale, original report	74.61
Gliding Vertex [10]	TPAMI	2020	ResNet-101	original reported protocol	75.02
R3Det [11]	AAAI	2021	ResNet-101	single-scale, original report	73.79
S2A-Net [30]	TGRS	2021	ResNet-50	single-scale, original report	74.12
GWD [22]	ICML	2021	R50-FPN	reproduced under our protocol	75.82
KLD [64]	NeurIPS	2021	R50-FPN	reproduced under our protocol	73.37
Oriented R-CNN [12]	ICCV	2021	ResNet-50-FPN	1024 input, original report	75.87
ReDet [15]	CVPR	2021	ReR50-ReFPN	single-scale, original report	76.25
ARC + Oriented R-CNN [12], [67]	ICCV	2023	ResNet-50-FPN	plug-in adaptive rotated convolution	81.77
DFDet [69]	TGRS	2024	ResNet-50	original reported protocol	74.71
ORSR [72]	TGRS	2024	ReResNet + ReFPN	single-scale, trainval, 1024 patches	78.03
DecoupleNet-D2 [76]	TGRS	2024	DecoupleNet-D2 + Oriented R-CNN	single-scale, original report	78.04
OrientedFormer [74]	TGRS	2024	ResNet-50	single-scale, original report	75.37
DDDet [73]	TGRS	2024	as reported	original reported protocol	79.41
PETDet [23]	TGRS	2024	ResNet-50-FPN	reproduced under our protocol	80.63
DRNet [63]	TCSVT	2025	ResNet-50	reproduced under our protocol	79.16
RS-vHeat [79]	ICCV	2025	vHeat-B + SCEs	reproduced under our protocol	82.94
AMMBA [75]	TGRS	2025	S2A-Net-based	original reported protocol	80.91
Baseline detector (YOLOv10-based) [66]	–	–	YOLOv10	our controlled single-scale protocol, 36 epochs	81.00
Ours (RAMMC + YOLOv10)	–	–	YOLOv10 + RAMMC	our controlled single-scale protocol, 36 epochs	**84.32**

D. Overall Comparison

We first report the fine-grained per-category comparison on FAIR1M-v2.0 in Table I. This table provides a detailed category-level evaluation across aircraft, ship, vehicle, court, and road subclasses. The results show that RAMMC achieves the best overall AP_{50} of 52.28 and obtains competitive or superior performance across most fine-grained categories, indicating its effectiveness for distinguishing visually similar remote-sensing objects under dense and arbitrary orientations.

We then compare our method with representative oriented object detectors and recent strong methods on DOTA-v1.0, as summarized in Table II. To further evaluate performance on the extended fine-grained benchmark, we also report comparison results on FAIR1M-v2.0 in Table III. These tables are intended to provide literature positioning across widely used remote-sensing benchmarks.

It should be noted that the results collected from recently published methods are not always obtained under exactly identical experimental protocols. Different papers may adopt different detector families, backbones, input resolutions, data-augmentation strategies, training schedules, cropping or slicing settings, and single-scale or multiscale testing protocols. Therefore, comparisons with methods such as DFDet [69], DecoupleNet [76], OrientedFormer [74], DDDet [73], DRNet [63], PETDet [23], and RS-vHeat [79] should be interpreted as representative external comparisons rather than strictly controlled same-setting ablations.

For controlled evaluation, we report same-setting comparisons between the baseline detector and the

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TABLE III

COMPARISON WITH THE SAME SET OF REPRESENTATIVE ORIENTED OBJECT DETECTION METHODS ON FAIR1M-V2.0

Method	Venue	Year	Backbone / Framework	Protocol	AP_{50} / mAP
Faster R-CNN [60]	NeurIPS	2015	as reported	original reported protocol	41.64
RetinaNet [61]	ICCV	2017	as reported	original reported protocol	26.07
FCOS [62]	ICCV	2019	as reported	original reported protocol	39.98
RoI Transformer [9]	CVPR	2019	as reported	original reported protocol	44.03
Gliding Vertex [10]	TPAMI	2020	as reported	original reported protocol	41.27
R3Det [11]	AAAI	2021	as reported	original reported protocol	36.04
S2A-Net [30]	TGRS	2021	as reported	original reported protocol	30.93
GWD [22]	ICML	2021	as reported	original reported protocol	31.48
KLD [64]	NeurIPS	2021	as reported	original reported protocol	32.12
Oriented R-CNN [12]	ICCV	2021	as reported	original reported protocol	43.90
ReDet [15]	CVPR	2021	as reported	original reported protocol	46.03
ARC + Oriented R-CNN [12], [67]	ICCV	2023	ResNet-50-FPN	reproduced under our protocol	50.14
DFDet [69]	TGRS	2024	ResNet-50	reproduced under our protocol	51.32
ORSR [72]	TGRS	2024	ReResNet + ReFPN	reproduced under our protocol	49.50
DecoupleNet-D2 [76]	TGRS	2024	DecoupleNet-D2 + Oriented R-CNN	reproduced under our protocol	46.64
OrientedFormer [74]	TGRS	2024	ResNet-50	reproduced under our protocol	49.43
DDDet [73]	TGRS	2024	as reported / reprod.	reproduced under our protocol	49.75
PETDet [23]	TGRS	2024	ResNet-50-FPN	original reported protocol	48.81
DRNet [63]	TCSVT	2025	ResNet-50	1× training schedule, original report	47.04
RS-vHeat [79]	ICCV	2025	vHeat-B + SCEs	original reported protocol	48.29
AMMBA [75]	TGRS	2025	S2A-Net-based	reproduced under our protocol	50.33
Baseline detector (YOLOv10-based) [66]	–	–	YOLOv10	our controlled single-scale protocol, 36 epochs	49.62
Ours (RAMMC + YOLOv10)	–	–	YOLOv10 + RAMMC	our controlled single-scale protocol, 36 epochs	**52.28**



Fig. 4. Qualitative comparison on FAIR1M-v2.0. (a) Original images, (b) baseline, (c) ground truth, and (d) ours. Prediction boxes are shown without labels for clearer visualization.

RAMMC-enhanced detector, and further provide component-wise ablations in Section IV. In the same-setting benchmark comparison, the two detectors share the same backbone, dataset split, optimization setting, and inference protocol. Under this setting, RAMMC improves the baseline detector from 49.62 to 52.28 AP_{50} on FAIR1M-v2.0 and from 81.00 to 84.32 AP_{50} on DOTA-v1.0. The method also achieves 47.31 AP_{50} on FAIR1M-v1.0. Qualitative comparisons are shown in Fig. 4, and the confusion matrices in Fig. 5 further confirm that the proposed module reduces category-level misclassifications when plugged into different baseline detectors.

These comparisons demonstrate that our design improves overall detection accuracy while maintaining strong robustness across representative remote-sensing benchmarks, even under challenging settings with heterogeneous object orientations and fine-grained categories.

E. Ablation Studies

We conduct comprehensive ablation experiments to evaluate the contribution of each component and to validate adaptability across different detection baselines. In this section, "Fea-Loss" denotes the proposed feature-level consistency regularizer

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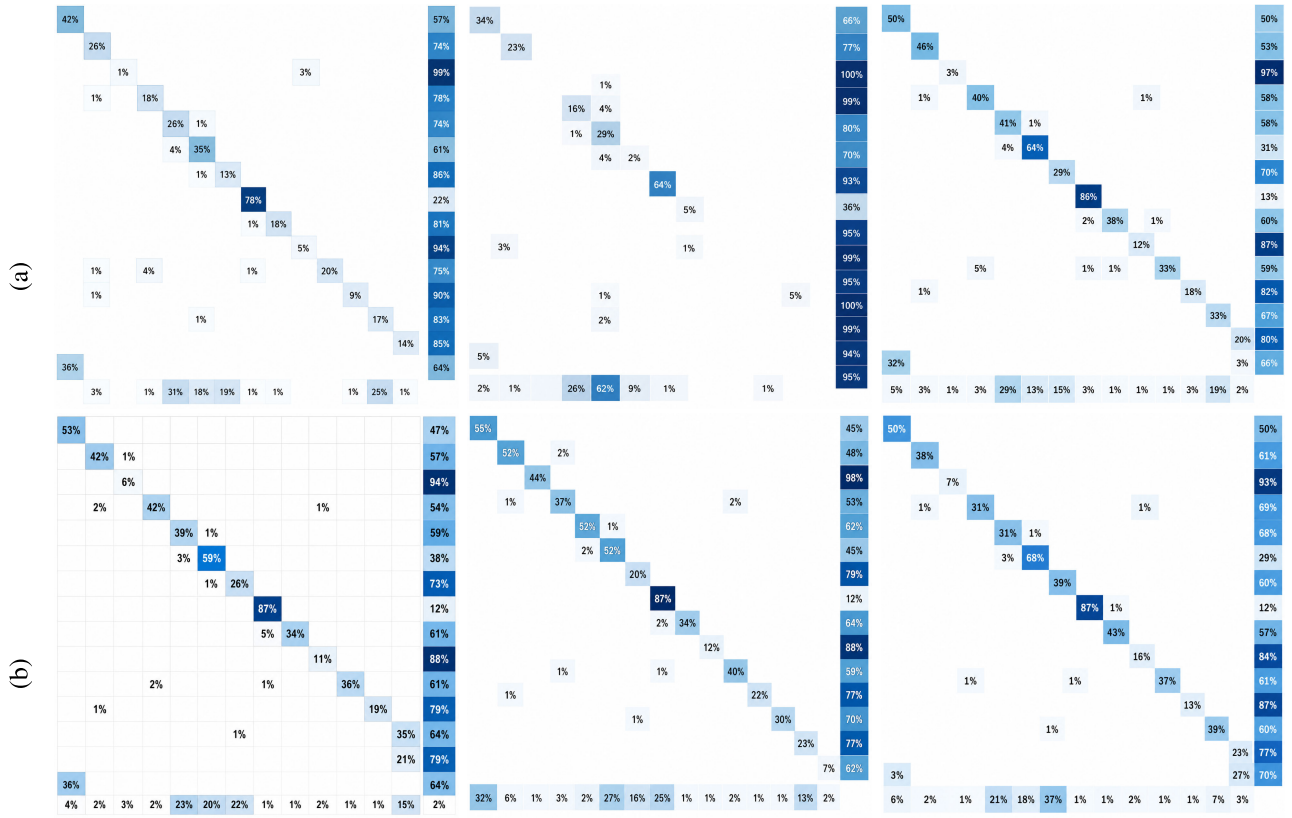


Fig. 5. Confusion matrices on FAIR1M-v2.0. Top row: base models; bottom row: models with the proposed RAMMC module. Columns correspond to Faster R-CNN, RoI Transformer, and Oriented R-CNN. Row/column labels are placed at the left/top to avoid captions under images. (a) Base model. (b) Add ours.

TABLE IV
EFFICIENCY ANALYSIS OF RAMMC ON DIFFERENT DETECTOR BACKBONES

Model	Params (M)	FLOPs (G)	Latency (ms)	Param Δ	FLOPs Δ	Latency Δ
YOLO11n-OBB	2.696	6.90	9.944	–	–	–
+ RAMMC	2.812	8.21	11.35	+4.3%	+19.0%	+14.1%
Oriented R-CNN	41.37	13.62	17.55	–	–	–
+ RAMMC	43.45	18.13	23.12	+5.0%	+33.1%	+31.7%
Rotated RetinaNet	36.64	11.80	15.11	–	–	–
+ RAMMC	38.72	15.16	19.43	+5.7%	+28.5%	+28.6%

TABLE V
ABLATION STUDY OF RAMMC AND REGULARIZERS ON THREE DATASETS

Dataset	RAMMC	Fea-Loss	Pre-Loss	AP_{50}
DOTA				81.00
DOTA			✓	81.76
DOTA	✓			82.56
DOTA	✓	✓		82.98
DOTA	✓		✓	82.63
DOTA	✓	✓	✓	**84.32**
FAIR1M				42.73
FAIR1M			✓	43.24
FAIR1M	✓			44.13
FAIR1M	✓	✓		45.09
FAIR1M	✓		✓	44.79
FAIR1M	✓	✓	✓	**47.31**
FAIR1M-v2.0				49.62
FAIR1M-v2.0			✓	49.80
FAIR1M-v2.0	✓			50.26
FAIR1M-v2.0	✓	✓		51.51
FAIR1M-v2.0	✓		✓	50.92
FAIR1M-v2.0	✓	✓	✓	**52.28**

$\mathcal{L}_{\text{eqv-feat}}$ (6), while "Pre-Loss" denotes the proposed global prediction-level consistency regularizer $\mathcal{L}_{\text{eqv-pred}}$ (7).

1) Componentwise Analysis: Efficiency results across different detector backbones are summarized in Table IV. Table V reports the incremental effects of each component. The baseline detector achieves 81.00, 42.73, and 49.62 AP_{50} on DOTA, FAIR1M, and FAIR1M-v2.0, respectively. Adding only the prediction-level loss yields moderate improvements, while RAMMC alone brings larger gains by explicitly modeling orientation. The feature-level regularizer further enhances equivariant consistency. Combining all three components provides the best results, improving AP_{50} by 3.32, 4.58, and 2.66 points across the three datasets.

Table VI reports the sensitivity of RAMMC to the main hyperparameters. When varying the number of mother kernels M from 2 to 6, the YOLO-based detector changes by within 0.24 AP_{50} points and Oriented R-CNN changes within 0.30 mAP points. When varying the channel reduction ratio β from 0.25 to 0.75, the variation is 0.23 AP_{50} points for the YOLO-based detector and 0.13 mAP points for Oriented R-CNN. When varying the consistency weight λ from 0.05 to 0.20, the variation is 0.08 AP_{50} points for the YOLO-based detector and 0.15 mAP points for Oriented R-CNN. These

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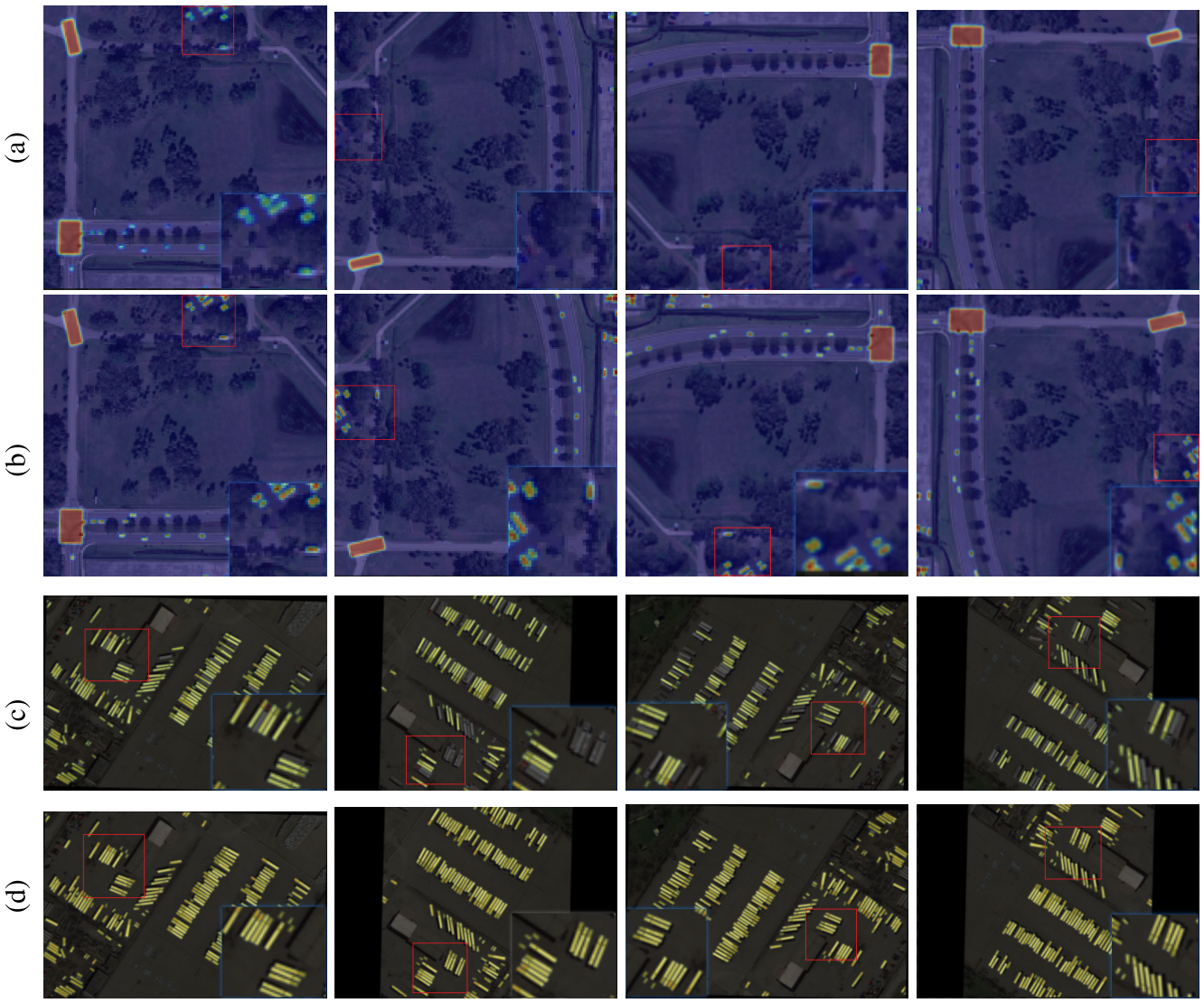


Fig. 6. Feature-activation visualizations under input rotations on FAIR1M-v1 and DOTA-v1. Columns correspond to rotation angles (0°, 90°, 180°, 270°). (a) Baseline on FAIR1M-v1. (b) Ours on FAIR1M-v1. (c) Baseline on DOTA-v1. (d) Ours on DOTA-v1.

TABLE VI

HYPERPARAMETER SENSITIVITY ANALYSIS OF RAMMC. ALL OTHER SETTINGS ARE FIXED. THE YOLO-BASED DETECTOR IS EVALUATED BY AP_{50}, WHILE ORIENTED R-CNN IS EVALUATED BY THE MMROTATE MAP METRIC UNDER THE SAME 5-EPOCH SENSITIVITY PROTOCOL

Detector	Hyperparameter	Setting 1	Setting 2	Setting 3
YOLO-based detector	M	2 / 38.67	4 / 38.81	6 / 38.91
YOLO-based detector	β	0.25 / 38.68	0.50 / 38.84	0.75 / 38.61
YOLO-based detector	λ	0.05 / 38.84	0.10 / 38.77	0.20 / 38.84
Oriented R-CNN	M	2 / 33.92	4 / 33.79	6 / 33.62
Oriented R-CNN	β	0.25 / 33.47	0.50 / 33.34	0.75 / 33.39
Oriented R-CNN	λ	0.05 / 33.79	0.10 / 33.64	0.20 / 33.73

results show that RAMMC is not highly sensitive to the tested hyperparameter choices. The default setting $M = 4$, $\beta = 0.5$, and $\lambda = 0.1$ is therefore used as a balanced configuration rather than as a finely tuned isolated optimum.

2) Cross-Dataset Visual Validation: Cross-dataset visual validation. Feature-activation visualizations under input rotations on FAIR1M-v1 and DOTA-v1 are shown in Fig. 6. To complement the quantitative ablation results, we further provide visual analysis from two perspectives.

First, Fig. 7 presents cross-dataset visualization of rotation consistency on DOTA-v1.0, FAIR1M-v1.0, and FAIR1M-v2.0. For each dataset, we compare the feature discrepancies before and after rotation alignment under different rotation angles. The gain maps show the discrepancy reduction after alignment, where larger values indicate better rotation consistency. As can be seen, our method consistently produces larger discrepancy reduction in target regions than the baseline across different rotation angles. The corresponding gain statistics and gain distributions further verify that this improvement is stable across datasets and samples, rather than arising from a few isolated cases.

Second, Fig. 8 provides cross-dataset visual ablation results to further validate the effects of different components. For each dataset, the response heatmaps are averaged over 128 samples and visualized with one shared color scale for fair comparison. Solid contours denote the target region, and dashed contours denote the surrounding background band. Here, Δt, Δb,

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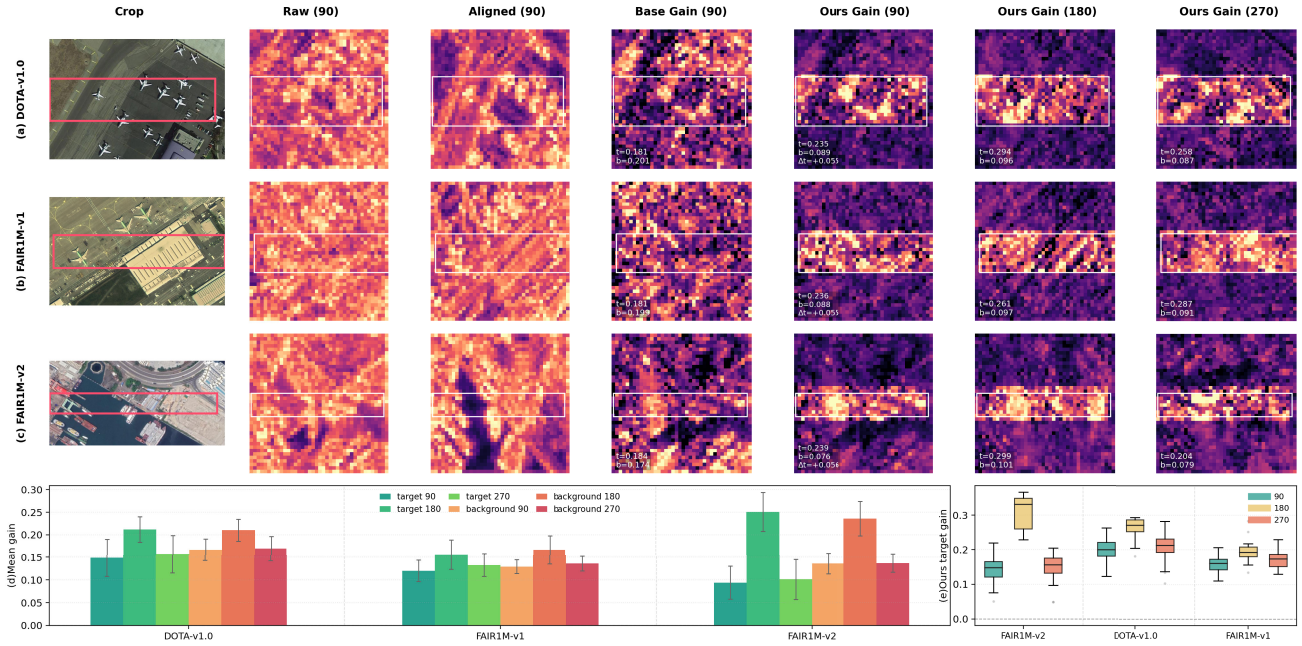


Fig. 7. Visualization of rotation consistency on DOTA-v1.0, FAIR1M-v1.0, and FAIR1M-v2.0. (a)–(c) Show example target patches and their feature-discrepancy maps before and after rotation alignment under different rotation angles. (d) Reports the mean gain statistics in target and background regions. (e) Distribution of target-region gains. Our method consistently yields larger discrepancy reduction in target regions, indicating improved rotation consistency across datasets.

Δsep, and Δstab denote the changes in target-region response, background-region response, foreground-background separation, and geometric stability, respectively, relative to the baseline, while the boxplots summarize the distribution of foreground–background separation across samples. The results show that the prediction-level regularizer mainly suppresses background interference, whereas RAMMC mainly enhances target responses and improves geometric stability. When combined with RAMMC, Fea-Loss further improves response consistency and stability, while Pre-Loss contributes more to foreground-background separation by suppressing background leakage. The full model achieves the best overall effect across all three datasets, yielding the strongest target activation, the weakest background interference, the clearest foreground-background separation, and the highest geometric stability.

3) Category-Level Evaluation: To further analyze detection behavior, we visualize confusion matrices in Fig. 5. The comparison confirms that RAMMC reduces interclass misclassifications when plugged into different baseline detectors, especially for visually similar categories in dense remote-sensing scenes.

4) Efficiency Analysis: Table IV further shows that RAMMC remains parameter-efficient across different detector families. The additional learnable parameters are 4.3%, 5.0%, and 5.7% for YOLO11n-OBB, Oriented R-CNN, and Rotated RetinaNet, respectively. The inference-time overhead is more visible because RAMMC explicitly evaluates orientation-structured responses, increasing latency by 14.1%, 31.7%, and 28.6% on the three representative models. These results indicate that RAMMC introduces a practical accuracy-efficiency tradeoff: it is compact in learnable parameters, while the extra computation is measurable and should be considered in deployment. Importantly, the consistency losses are disabled during inference, so the reported latency reflects only the deployed RAMMC adaptor.

5) Training-Time Overhead of Consistency Regularization: The feature-level equivariance loss requires a paired rotated view of the input image in order to compare the feature representation before and after geometric alignment. Therefore, during training, the network processes both the original image and its rotated counterpart for the consistency branch. This introduces additional training-time computation and may reduce training throughput compared with the baseline detector. The standard detection loss is kept as the primary supervised objective, while the feature-level and prediction-level consistency losses are used as auxiliary regularizers

$$\mathcal{L} = \mathcal{L}_{\text{det}} + \lambda_f \mathcal{L}_{\text{eqv-feat}} + \lambda_p \mathcal{L}_{\text{eqv-pred}}.$$

At inference time, the detector takes only a single input image, and both consistency losses are disabled. Therefore, this paired-view cost is an offline training-only overhead and does not introduce additional inference-time computation. The deployment latency reported in Table IV thus reflects only the deployed RAMMC adaptor.

To provide feature-level evidence, we further conduct a feature visualization experiment, as shown in Fig. 6. For each image, we rotate the input by 0°, 90°, 180°, and 270°, and visualize the aggregated high-level feature responses as an activation heatmap overlaid on the image. The baseline detector exhibits unstable or missing activations under rotation, whereas our method produces more concentrated and

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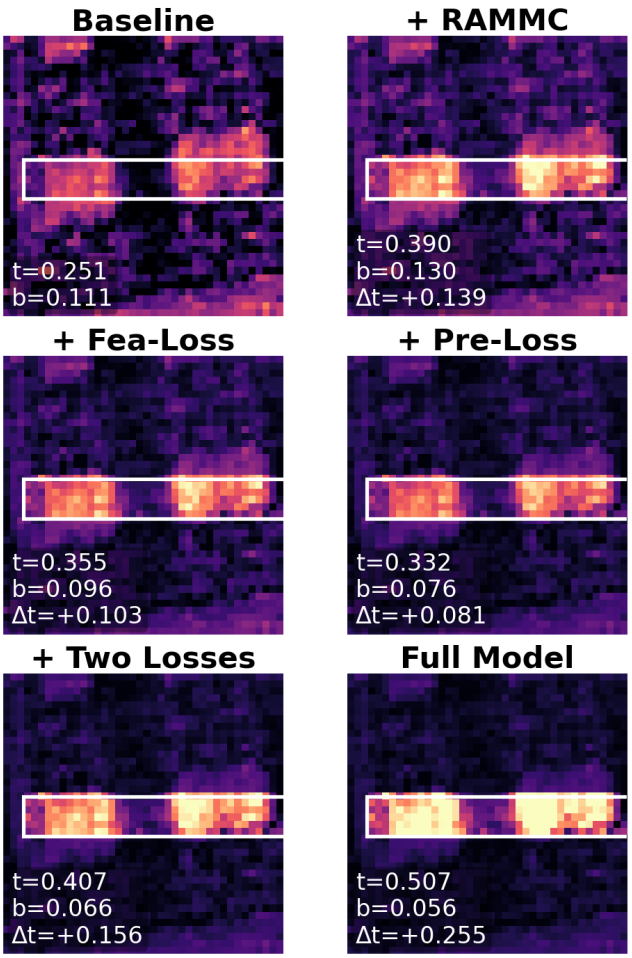


Fig. 8. Visual ablation of different components on DOTA-v1.0, FAIR1M-v1.0, and FAIR1M-v2.0. Averaged response heatmaps and separation distributions show that +Pre mainly suppresses background interference, +RAMMC mainly enhances target responses and improves geometric stability, and the full model achieves the best overall effect.

rotation-consistent responses, especially on small and densely distributed objects.

For the baseline detector, the highlighted region becomes significantly weakened or even disappears after rotation, indicating a clear drop in detection capability under geometric transformation. In contrast, the proposed method preserves strong and spatially aligned activations on the same region across all rotation angles, demonstrating that our model successfully captures rotation-consistent feature patterns rather than relying on orientation-specific responses.

6) Summary: The ablation results collectively confirm three key findings: 1) RAMMC consistently improves detection accuracy across diverse datasets and detector architectures, highlighting its generality; 2) the two regularizers play complementary roles, with the feature-level term enhancing equivariance and the prediction-level term stabilizing angle regression; and 3) these accuracy gains are achieved while maintaining inference latency in a practical range.

V. Conclusion

We presented RAMMC, an implementation-aware rotation-adaptive module for remote sensing object detection. Unlike heuristic plug-ins, RAMMC encodes rotation geometry through rotated kernel banks and orientation-aware aggregation, complemented by feature-level and prediction-level consistency regularizers. The revised analysis shows that its group-pooling branch realizes a Reynolds averaging operator and that the remaining equivariance defect can be decomposed into finite kernel-rotation, gating, attention, and decoding-related residuals under stated assumptions.

This conditional analysis avoids claiming exact end-to-end equivariance for the complete detector and instead provides practical guidance for controlling rotation-induced mismatch in the implemented network. Extensive experiments across multiple benchmarks demonstrate consistent gains (e.g., 84.32 mAP on DOTA-v1.0, 47.31 mAP on FAIR1M-v1, and 52.28 mAP on FAIR1M-v2.0), validating the generality of the framework.

Overall, this work shows that embedding *mathematically principled geometric priors* into detection networks is both feasible and effective. Beyond accuracy improvements, RAMMC highlights a pathway for bridging theory and practice in high-resolution aerial imagery. Future work will investigate real-time deployment, multimodal integration, and extensions to broader equivariant structures in vision.

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